

VADSO, Norway

WMO: 010880

Lat: 70.065N

Lon: 29.835E

Elev: 128

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.5	7.1	-0.9	5.3	4.5	2.8	6.4	8.1	38.8	28.2	34.9	27.0	12.9	280	0.870

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	67.7	58.9	63.9	56.7	60.5	54.8	59.9	66.2	57.5	62.4	55.3	59.8	8.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.7	68.9	63.2	54.7	64.0	60.6	52.7	59.3	58.0	26.4	66.2	24.8	62.7	23.4	59.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.9	28.1	24.9	-2.8	76.1	4.4	5.0	-5.9	79.7	-8.5	82.7	-11.0	85.5	-14.2	89.1		
(5)			-1.5	62.7	5.1	3.3	-5.2	65.1	-8.2	67.0	-11.1	68.9	-14.8	71.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	21.5	20.3	24.4	30.4	39.3	46.2	51.9	50.6	44.6	35.6	27.9	25.7
	DBStd	12.49	6.65	7.87	6.54	5.80	5.80	5.88	5.61	5.04	5.12	5.92	6.21	5.88
	HDD50	5710	882	831	793	588	336	147	40	53	174	448	664	754
	HDD65	10967	1347	1251	1258	1038	797	564	408	447	611	913	1114	1219
	CDD50	220	0	0	0	0	4	33	99	71	13	0	0	0
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDH74	19	0	0	0	0	0	0	15	4	0	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	12.6	14.8	14.9	15.0	12.3	11.8	12.3	10.7	9.6	10.8	12.0	13.4	14.3
Precipitation	PrecAvg	21.8	1.8	1.7	1.4	1.1	1.3	1.8	2.3	2.4	2.1	2.6	1.7	1.7
	PrecMax	26.5	2.9	2.4	2.5	2.2	2.3	3.7	5.2	6.0	3.4	4.3	2.8	2.6
	PrecMin	18.1	0.9	0.8	0.4	0.4	0.7	0.3	0.5	0.9	1.2	0.9	0.5	0.7
	PrecStd	2.2	0.4	0.4	0.5	0.4	0.5	0.9	1.3	1.2	0.6	0.8	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.1	37.8	39.5	47.5	61.1	68.5	75.9	71.3	61.5	50.3	42.8	39.2
		MCWB	36.8	35.5	35.8	42.1	51.0	58.9	63.3	62.3	55.4	46.6	40.9	37.1
	2%	DB	35.9	35.8	37.5	42.8	55.9	63.9	69.5	65.0	57.5	47.9	39.6	37.5
		MCWB	33.5	33.5	34.6	38.8	47.9	55.2	59.9	57.8	52.6	45.3	37.8	35.6
	5%	DB	33.6	33.8	35.7	39.9	52.0	59.3	65.2	61.8	55.1	45.8	37.7	35.9
		MCWB	31.6	31.8	33.1	37.1	45.5	52.5	57.9	56.0	51.2	43.7	35.7	33.8
	10%	DB	31.5	31.4	33.8	38.2	48.5	55.3	61.1	59.0	52.4	43.9	35.9	34.2
		MCWB	29.9	29.7	31.3	35.7	43.3	49.6	55.9	54.4	49.0	41.9	34.2	32.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	36.0	36.3	42.1	52.0	60.4	64.2	62.8	56.3	46.9	41.0	37.3
		MCDB	38.7	37.7	38.9	46.4	58.7	67.3	74.0	69.3	60.5	49.1	42.2	38.8
	2%	WB	33.8	33.9	34.8	39.1	48.2	56.0	60.8	58.9	53.2	45.4	38.0	35.8
		MCDB	35.5	35.6	37.2	41.9	55.2	62.2	68.3	64.1	56.6	47.4	39.4	37.4
	5%	WB	31.7	31.9	33.2	37.4	46.1	53.0	58.4	56.7	51.3	43.7	35.9	34.1
		MCDB	33.2	33.7	35.4	39.6	51.5	59.0	64.1	60.7	54.6	45.4	37.4	35.5
	10%	WB	29.8	29.6	31.6	36.1	43.8	50.3	56.0	54.8	49.7	41.9	34.3	32.3
		MCDB	31.4	31.2	33.6	37.9	48.0	54.8	60.8	58.4	52.2	43.6	35.8	33.8
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.0	8.6	9.5	10.5	10.1	10.6	11.0	9.6	6.9	7.7	8.0
		MCDBR	10.0	8.7	7.5	9.1	17.1	16.5	18.5	15.9	12.8	6.8	7.1	7.0
		MCWBR	9.3	8.0	6.6	6.9	10.7	10.1	10.3	9.8	9.0	6.0	6.7	6.8
	5% WB	MCDBR	10.0	8.7	7.3	8.2	16.2	16.2	17.1	14.3	11.8	6.2	7.0	7.2
		MCWBR	9.5	8.4	6.6	6.4	10.3	10.2	10.4	9.5	8.6	5.7	6.8	7.1
Clear-Sky Solar Irradiance	taub	N/A	0.263	0.312	0.338	0.360	0.352	0.368	0.351	0.317	0.275	N/A	N/A	
	taud	N/A	2.038	2.121	2.192	2.317	2.441	2.430	2.457	2.497	2.180	N/A	N/A	
	Ebn at Noon	0	136	212	249	259	267	257	247	224	133	0	0	
	Edh at Noon	0	13	25	32	32	30	29	25	18	11	0	0	
All-Sky Solar Radiation	RadAvg	0	91	467	1001	1361	1509	1375	935	494	138	0	0	
	RadStd	0	11	32	76	144	114	134	92	43	16	0	0	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (16 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page